

Predicting Hospital Readmissions



A Machine Learning Approach to Proactive Risk Management

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What is Readmissions?

According to CMS.gov¹, unplanned readmissions that happen within 30 days of discharge from the index (that is, initial) admission. The readmission measures include patients who are readmitted to the same hospital, or another applicable acute care hospital, no matter the principal diagnosis. The measures exclude some planned readmissions.

CMS refers to the Center for Medicare & Medicaid Services, a U.S federal government agency that provides healthcare coverage to more than 160 million people

¹ <https://www.cms.gov/medicare/quality/value-based-programs/hospital-readmissions>

Why Readmissions Matter



3%

Medicare Penalty

Under CMS HRRP for excess readmissions



18.2%

Readmission Rate

vs 15.5% national average



\$8.5M

Annual Cost Burden

\$2.4M penalties + \$6.1M preventable costs

Key Gap: No consistent way to identify high-risk patients at discharge — leading to reactive rather than proactive care.

PROJECT GOAL

Build a machine learning model to transform how Healthbridge manages readmission risk



Predict Risk



Score Patients



Target Interventions



Drive Outcomes

Outcome: Better predictions → proactive care → lower costs

UCI Diabetes 130-US Hospitals Dataset

130

Hospitals

US facilities represented

101,766

Raw records

Original dataset size

69,990

Clean records

After deduplication & filtering

9%

Positive class

<30-day readmissions

DATA PREPARATION

How raw hospital records became a balanced, model-ready dataset



01 Cleaning & Dedup



02 Feature Engineering



03 Scaling & Splitting

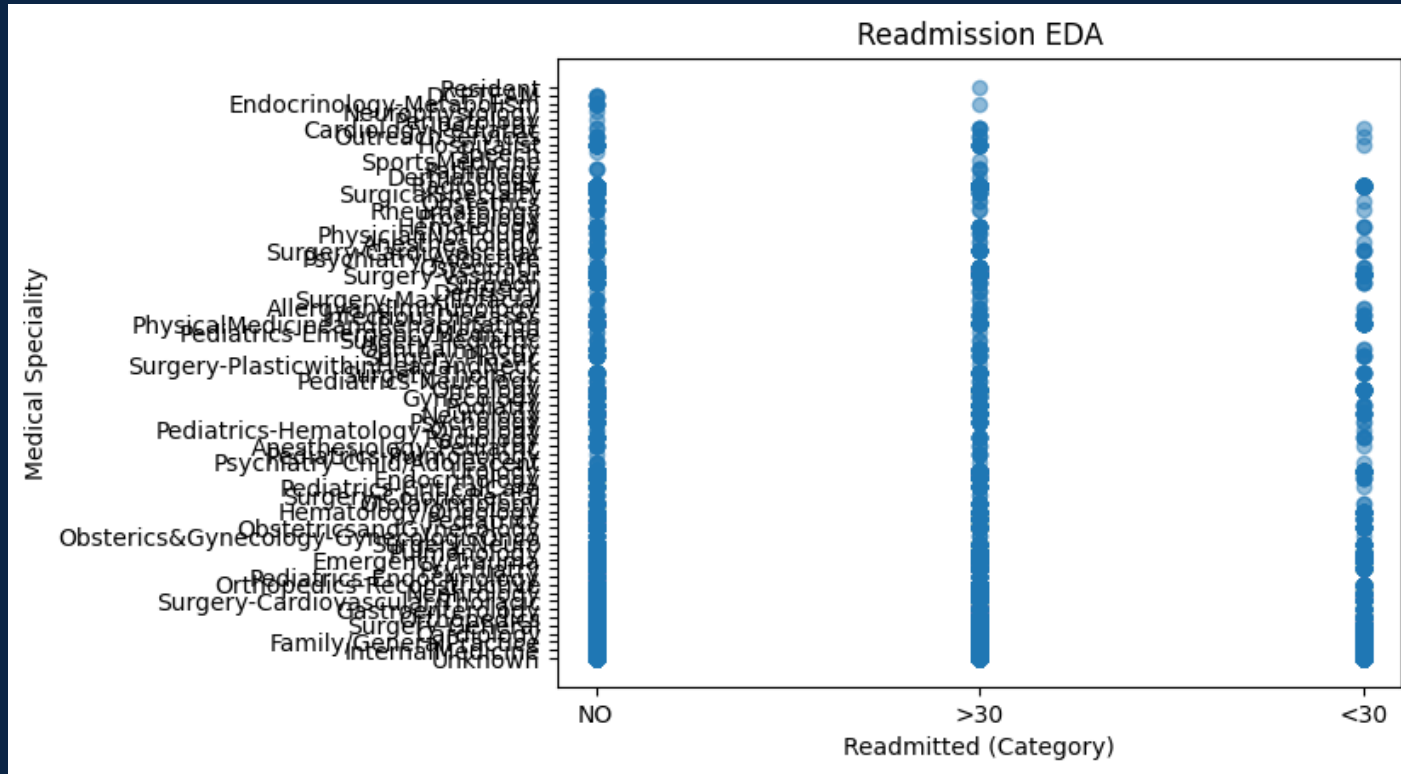


04 SMOTE Balancing

Result: 101,928 balanced training rows × 23 features | 13,998 natural test rows — ready for modeling

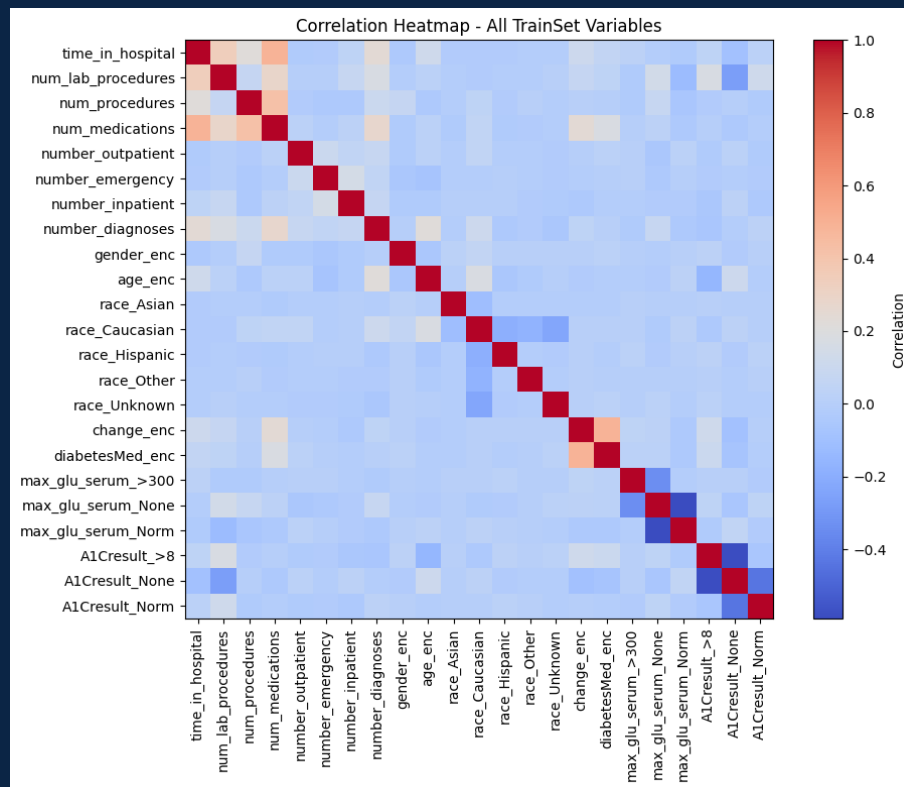
Exploratory Data Analysis

Target variable (Readmitted) - Raw Data



Exploratory Data Analysis – Contd.

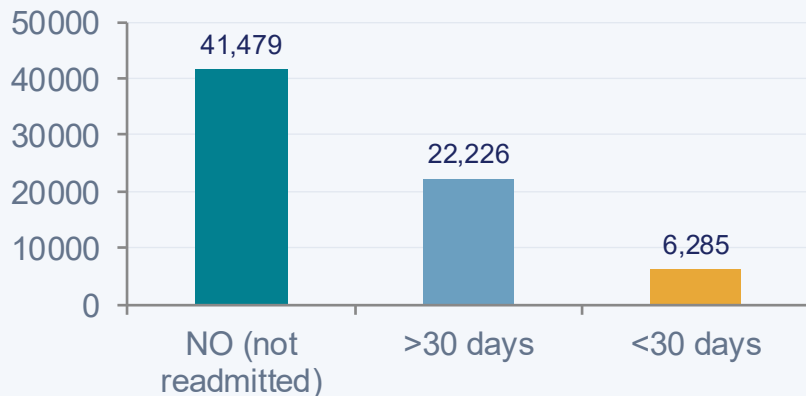
Correlation Heatmap



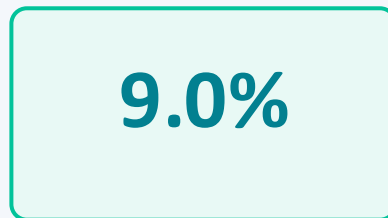
Target Variable — Readmission Descriptive Statistics

Original Target Variable (readmitted)

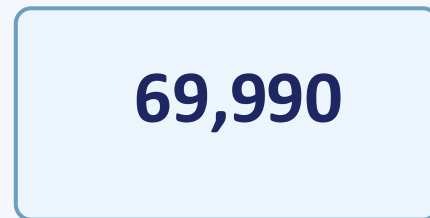
Statistic	Value
Count	69,990
Unique Categories	3
Most Frequent (Top)	NO
Top Frequency	41,479



Binary Target Variable (readmitted_30)

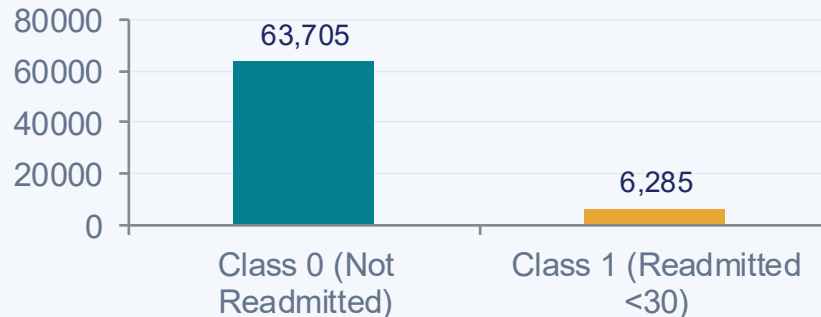


Positive Class Rate



Total Records

Binary Class Distribution



Class imbalance: 91% Class 0 vs 9% Class 1 — addressed during modeling

Three Algorithms Evaluated



Logistic Regression

Accuracy

61.2%

ROC-AUC

0.577

Low overall accuracy; not strong for clinical deployment



Random Forest

Accuracy

89.7%

ROC-AUC

0.568

Highest accuracy; clear feature importance ranking



XGBoost ★ Best

Accuracy

71.4%

ROC-AUC

0.591

Best ROC-AUC; balanced trade-off between precision and recall

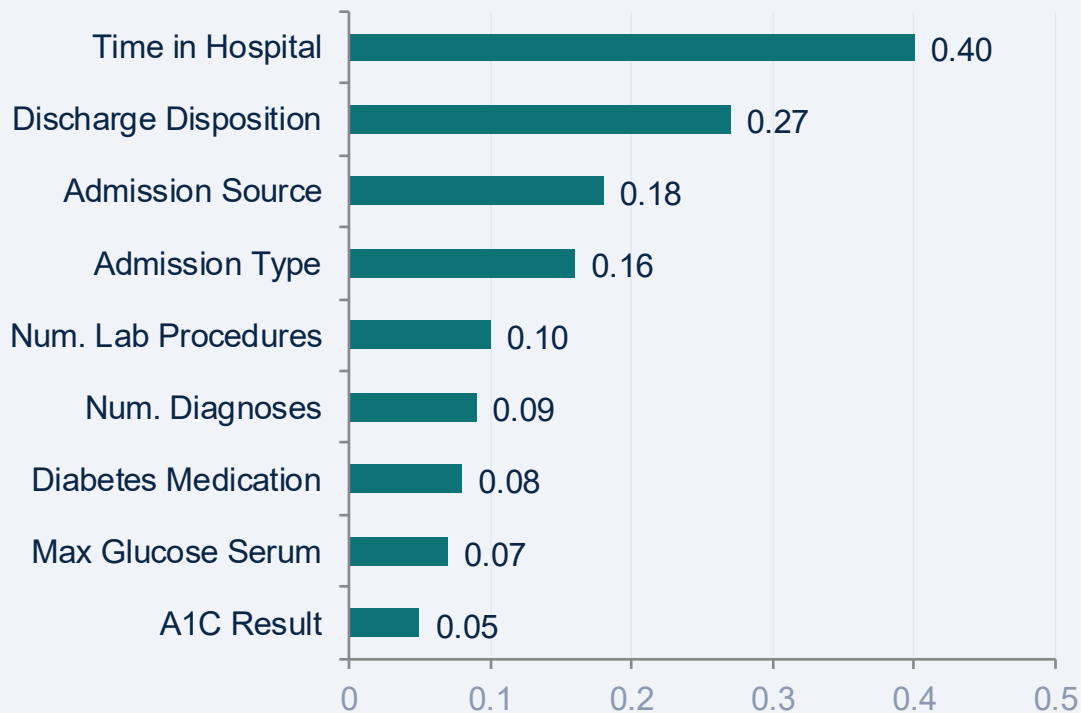
RANDOM FOREST KEY DRIVERS

Top Predictive Features (Random Forest Importance)



XGBOOST KEY DRIVERS

What Predicts Readmission?



RECOMMENDATIONS

Clinical action items and path forward

CLINICAL RECOMMENDATIONS



Engage multidisciplinary teams early



Strengthen care transitions



Use model risk scores to prioritize post-discharge



Address high-risk demographics

TECHNICAL NEXT STEPS



Request additional variables like comorbidity indices



Try cost-sensitive learning using deep learning models



Add early stopping to XGBoost to address detected overfitting



Integrate real-time EHR data feeds for live risk scoring at the point of discharge

CONCLUSION

**Better
predictions**

Better care

Lower cost

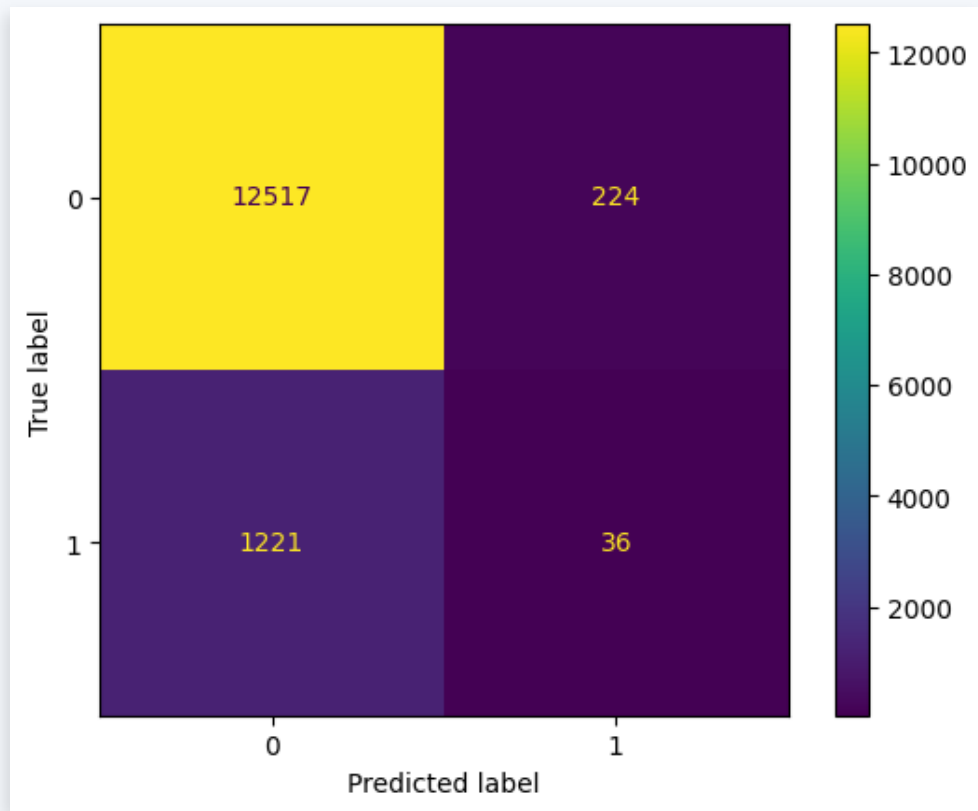
Citation and Attribution

Claude (Anthropic). (2025). AI coding assistance, analytical collaboration, PowerPoint compilation & design. Claude Sonnet 4, claude.ai.

Portions of this report were reviewed and refined with assistance from Microsoft Copilot, an AI companion developed by Microsoft.”

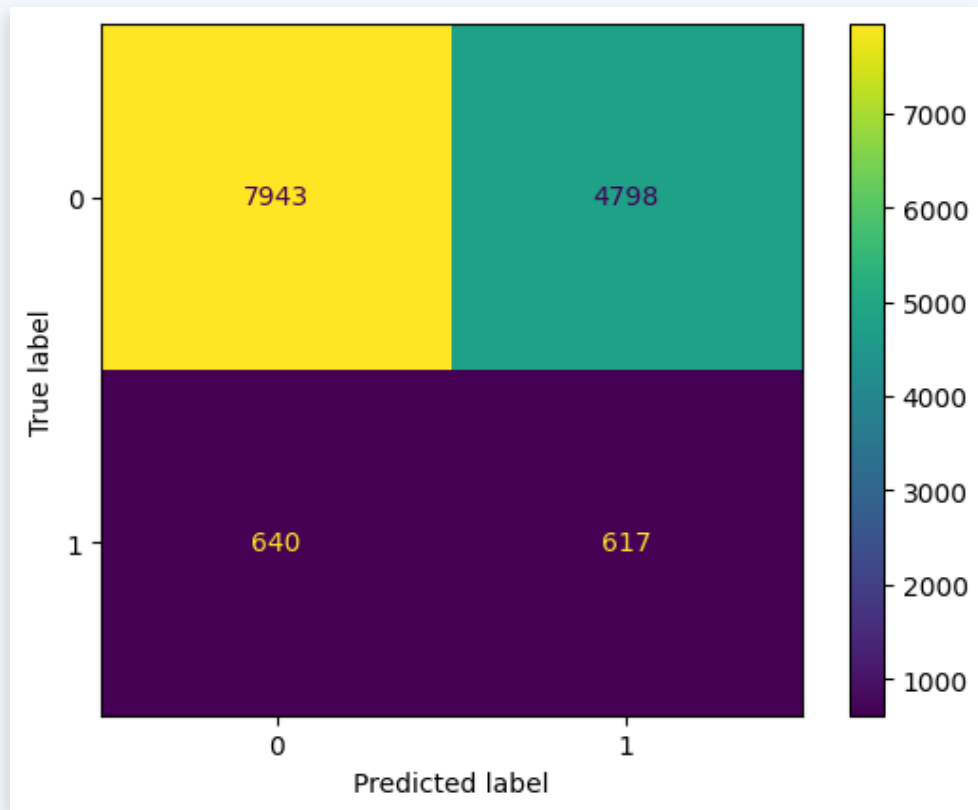
APPENDIX

Random Forest — Confusion Matrix



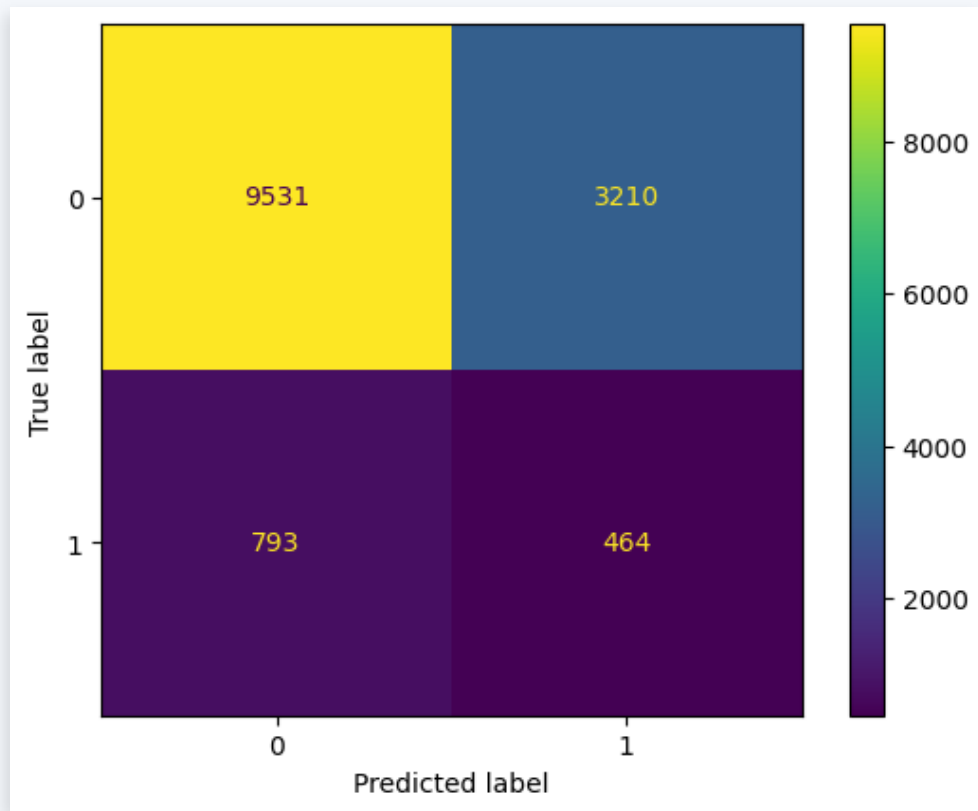
True label → Rows | Predicted label → Columns

Logistic Regression — Confusion Matrix



True label → Rows | Predicted label → Columns

XGBoost — Confusion Matrix



True label → Rows | Predicted label → Columns

Classification Report — Model Comparison

Model	Class	Precision	Recall	F1-Score	Support
Random Forest	0 (No Readmit)	0.9111	0.9824	0.9454	12,741
	1 (Readmitted)	0.1385	0.0286	0.0475	1,257
	Accuracy			0.8968	13,998
	Weighted Avg	0.8417	0.8968	0.8648	13,998
Logistic Regression	0 (No Readmit)	0.93	0.62	0.74	12,741
	1 (Readmitted)	0.11	0.49	0.18	1,257
	Accuracy			0.61	13,998
	Weighted Avg	0.85	0.61	0.69	13,998
XGBoost	0 (No Readmit)	0.92	0.75	0.83	12,741
	1 (Readmitted)	0.13	0.37	0.19	1,257
	Accuracy			0.71	13,998
	Weighted Avg	0.85	0.71	0.77	13,998

Class 0 = Not Readmitted | Class 1 = Readmitted | Support = number of test samples